ENTERPRISE CREDIT DECISIONING STRATEGY SIMULATOR

Module 1: Synthetic Application & Risk Modeling Engine

Validation Summary (Board-Ready)

1. EXECUTIVE SUMMARY

The Module 1 Synthetic Application & Risk Modeling Engine has been successfully validated as a **portfolio-grade, deterministic, and product-aware simulation framework** designed to emulate real-world credit decisioning environments without exposing sensitive data.

This validation confirms that the engine:

- Produces **stable, reproducible populations** using deterministic seeding tied to `population_id`
- Accurately reflects **configured portfolio composition** across product types and credit score segments
- Generates **behaviorally realistic borrower profiles**, preserving overlap across risk dimensions
- Maintains **coherent financial relationships** between income, exposure, payment burden, and risk
- Demonstrates a **directionally correct and explainable risk model**, where Probability of Default (PD) and Expected Loss (EL) respond appropriately to borrower and product characteristics

A critical calibration improvement—focused on mortgage exposure and affordability—was successfully implemented and validated, eliminating previously observed unrealistic payment burden distortions.

Overall, Module 1 has achieved its objective:

To function as a governed, explainable, and portfolio-realistic simulation engine suitable for pre-production credit strategy testing.

2. VALIDATION APPROACH OVERVIEW

Validation was conducted using a structured, multi-layer QA framework embedded directly within the SQL code (Section 9). This framework evaluates the simulation across five dimensions:

1. **Structural Integrity**
2. **Distribution Alignment**
3. **Behavioral Realism**
4. **Risk Model Coherence**
5. **Edge Case & Overlap Validation**

Each dimension includes both:

- Aggregate statistical checks
- Targeted "spot check" queries for interpretability and anomaly detection

3. PORTFOLIO STRUCTURE VALIDATION

3.1 Product Mix Alignment

Objective: Confirm that generated product distribution matches configured assumptions.

Observed Output

Product Type	Applications	% of Total
Mortgage	15,065	30.13%
Revolving Line	12,455	24.91%
Auto Loan	9,946	19.89%
Unsecured Personal Loan	7,572	15.14%
HELOC	4,962	9.92%

Assessment

- Target mix: **30 / 25 / 20 / 15 / 10**
- Observed deviation: **< 0.2% across all categories**

Conclusion

Product assignment logic is **highly precise and stable**, confirming correct implementation of cumulative probability allocation.

3.2 Score Band Distribution

Score Band	Applications	% of Total
Super Prime	8,984	17.97%
Prime	13,987	27.97%
Near Prime	12,073	24.15%
Subprime	8,902	17.80%
Deep Subprime	6,054	12.11%

Assessment

- Balanced distribution across all bands
- No collapse into extreme segments
- Adequate density for downstream analytics

Conclusion

The engine produces a **well-shaped portfolio**, enabling meaningful segmentation and risk analysis.

4. DETERMINISTIC GENERATION VALIDATION

4.1 Reproducibility Design

The engine replaces session-based randomness (`random()`) with deterministic hashing:

- MD5(application_id + population_id + deterministic_key_suffix)
- Converted to uniform(0,1) values

4.2 Validation Outcome

- Identical inputs produce **identical outputs across reruns**
- Enables:
  - Scenario comparison
  - Stable QA results
  - Debug traceability

Conclusion

The system behaves as a **controlled simulation environment**, not a stochastic generator.

5. VARIABLE DISTRIBUTION VALIDATION

5.1 Key Portfolio Statistics (Selected)

Variable	Median	Mean	P75	Max
Annual Income	~$118K	~$122K	~$162K	~$239K
Requested Amount	~$47K	~$135K	~$183K	~$1.15M
PTI	~11.3%	~16.5%	~25.9%	~132%
Estimated PD	~8.7%	~16.7%	~33.6%	45%

All statistics are based on a deterministic population of 50,000 synthetic applications.

5.2 Observations

- Distributions show:
  - **Healthy spread** (not compressed)
  - Realistic skew (right-tailed exposure)
  - Controlled extremes (no runaway values)

Conclusion

Variables exhibit **portfolio-level coherence**, not independent random behavior.

6. PRODUCT-AWARE BEHAVIOR VALIDATION

6.1 Key Findings

Mortgage

- Large balances (avg ~$373K)
- Moderate PTI (~0.33 avg)
- Lower APR (~8–10%)

Revolving

- Smaller balances (~$17K avg)
- Higher APR (~24–25%)
- Lower PTI

Auto

- Mid-range balances (~$43K)
- Structured amortization behavior

Insight

Product structure meaningfully impacts affordability, pricing, and risk.

7. SCORE-BAND BEHAVIOR VALIDATION

7.1 Observed Trends

Metric	Strong Bands	Weak Bands
Income	Higher	Lower
DTI	Lower	Higher
Utilization	Lower	Higher
Delinquencies	Minimal	Elevated
PD	Low	High

Example

- Super Prime PD ≈ 1–2%
- Deep Subprime PD ≈ 45%

Conclusion

Score bands behave **directionally correct across all dimensions**, confirming model coherence.

8. RISK MODEL VALIDATION

8.1 PD Gradient Validation

Example (Auto Loans):

Band	Avg PD
Super Prime	0.0119
Prime	0.0361
Near Prime	0.1200
Subprime	0.3707
Deep Subprime	0.4494

Assessment

- Clean monotonic progression
- No inversions
- No mid-band compression

Known Limitation

- **Cap crowding** observed near 0.45 in weakest segments

Conclusion

PD model is **functionally correct**, with refinement opportunity identified.

8.2 PTI → PD Relationship

PTI Bucket	Avg PD
<5%	Lowest
5–9%	↑
10–14%	↑
15–19%	↑
20%+	Highest

Interpretation

- Increasing affordability burden → increasing risk

Conclusion

PTI functions correctly as an **affordability bridge into risk**.

9. MORTGAGE CALIBRATION VALIDATION

9.1 Original Issue

- Mortgage exposure scaled too aggressively with income
- Resulted in:
  - Unrealistic PTI (>50%)
  - Distorted risk signals

9.2 Implemented Fix

- Introduced **income-based scaling constraints**
- Balanced:
  - Exposure size
  - Payment affordability

9.3 Post-Fix Evidence

Example validated records:

- $450K mortgage
- PTI ≈ 14–15%
- Income ≈ $230K

Conclusion

Mortgage behavior is now **realistic, controlled, and portfolio-consistent**.

10. MIXED-SIGNAL REALISM VALIDATION

10.1 Strong Credit + High PTI

- Exists but rare
- Demonstrates flexibility

10.2 Weak Credit + Low PTI

- Exists but constrained
- Prevents deterministic bias

Why This Matters

Real portfolios contain:

- Overextended strong borrowers
- Conservative weak borrowers

Conclusion

The engine preserves **behavioral overlap**, avoiding artificial segmentation.

11. EDGE CASE VALIDATION

11.1 Large Exposure + Low PTI

- Present and realistic
- Indicates proper scaling

11.2 High PTI Outliers

- Exist but not dominant
- Reflect real-world tail risk

11.3 Amount-to-Income Ratios

- High values exist but remain controlled

Conclusion

Tail behavior is **present but governed**, supporting realism without instability.

12. OVERALL VALIDATION CONCLUSION

Module 1 successfully demonstrates:

✓ Deterministic simulation framework

✓ Portfolio-level realism

✓ Product-aware modeling

✓ Coherent affordability and risk relationships

✓ Explainable and auditable outputs

This positions the engine as:

A pre-production decision testing platform, not just a synthetic dataset generator

13. PLANNED FUTURE ENHANCEMENTS (V2.0)

13.1 Mortgage Tail Refinement

- Further smooth upper-bound affordability behavior
- Introduce soft constraints for extreme income/exposure ratios

13.2 PD Cap Smoothing

- Replace hard cap with:
  - logistic tapering
  - segment-aware ceilings

13.3 Revolving Payment Calibration

- Incorporate:
  - utilization sensitivity
  - APR-driven payment variability

13.4 Enhanced Validation Documentation

- Expand:
  - scenario comparison framework
  - regression testing narratives
  - sensitivity analysis outputs

14. FINAL STATEMENT

Module 1 has achieved its intended design goal:

To simulate a governed, explainable, and portfolio-realistic credit decisioning environment capable of supporting strategy testing at scale.

The system is:

- Stable
- Interpretable
- Extensible

and ready to serve as the foundation for advanced decisioning modules.